Programming for GIS, Clark University

Automating Map Comparison using the Mean Absolute Deviation

An ArcGIS script tool using Python

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Introduction

Mean Absolute Deviation (MAD) measures the average of the absolute differences between two variables, designated by X and Y. MAD is applicable for situations where X and Y measure or represent the same phenomenon over space or time. Often this type of comparison is done insufficiently using regression analysis. MAD has several desirable properties over regression analysis; it is unbounded, symmetric and is expressed in the same units as the deviations that it measures. Scientists should use MAD in order to compare datasets of the same continuous variable. MAD can be utilized to test the reliability of a dataset against another measuring the same variable by indicating how much one deviates from the other. MAD can also be used to test model predictions against data representing known reality. This paper describes a program created to automate MAD analysis of spatial data within the ArcMap GIS using the Python programming language and ArcPy.

Uses of MAD in spatial analysis and GIS have focused on the analysis of raster data. This program revises the MAD raster equation in order to analyze variables assigned to vector polygons. The program checks the projection of input spatial data to insure it has an equal-area projection. If the file does not have an equal-area the program projects it into a default equal-area sinusoidal projection. MAD consists of three components: Quantity, Stratum, and Element (figure 1). Quantity is the absolute difference between the mean of X and the mean of Y, indicating the disagreement between them. Stratum-level disagreement is positive when at least one stratum has a positive residual and at least one other stratum has a negative residual. Element-level component of deviation is positive when one polygon has a positive residual and another polygon within the same stratum has a negative residual.

$$W = \frac{A_e}{A_{e^0}}$$

$$Ds_e = (Ys_e - Xs_e)W$$

$$\text{Element Component} = \frac{\sum_{s=1}^{S} \sum_{e=1}^{E} |Ds_e|}{N}$$

$$\text{Stratum Component} = \frac{\sum_{s=1}^{S} \left|\sum_{e=1}^{E} Ds_e\right|}{N}$$

$$\text{Quantity Component} = \frac{\left|\sum_{s=1}^{S} \sum_{e=1}^{E} Ds_e\right|}{N}$$

$$\text{Total} = \textit{Element} + \textit{Stratum} + \textit{Quantity}$$

W is the weight
A_e is the area for the current element e
A_{e^0} is the area of the smallest element e
Xs_e is the X value in stratum s for the element e
Ys_e is the Y value in stratum s for the element e

Ds_e is the Ys_e minus Xs_e deviation in stratum s for the element e
S is the number of strata
N is the number of elements

Figure 1. The MAD equation utilized in this program. This equation has been modified from its raster application in order to use polygon area for weighting each Element.

Data

Two spatial data sets were utilized for testing the MAD program. The first is a manually created shapefile of eight features containing arbitrary values for X, Y and Strata assignment (figure 2). These data were designed so that MAD and its components can be calculated manually in order to design and test the functionality of the program. Manually calculating MAD and its components ensured area-weight assignments could operate properly and the program produced accurate results. The second data set utilized in testing was a shapefile containing 2010 and 2013 population data of all states in the conterminous U.S. (figure 3). The data was acquired from the U.S. Census Bureau. Each state polygon designates an Element and the regions of the U.S. delineate the Strata. Population in 2010 was the X field and population in 2013 was the Y field. This population dataset was utilized to test the projection function included in the MAD program. Transforming, projecting and testing the shapefile using various datums and projections ensured the program would produce the correct area-weights for each element, correctly project the input shapefile into an equal-area projection, and generate necessary error messages. Read me file is included with project data.

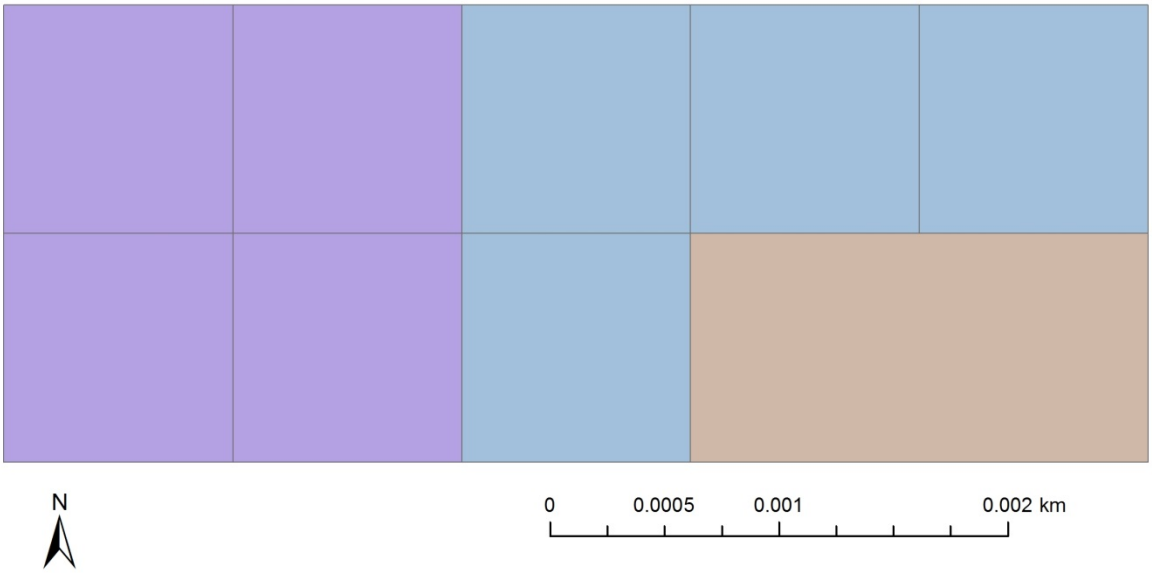

Figure 2. Shapefile created for testing MAD program. Each polygon (delineated by grey lines) are an Element and each color designates a Strata.

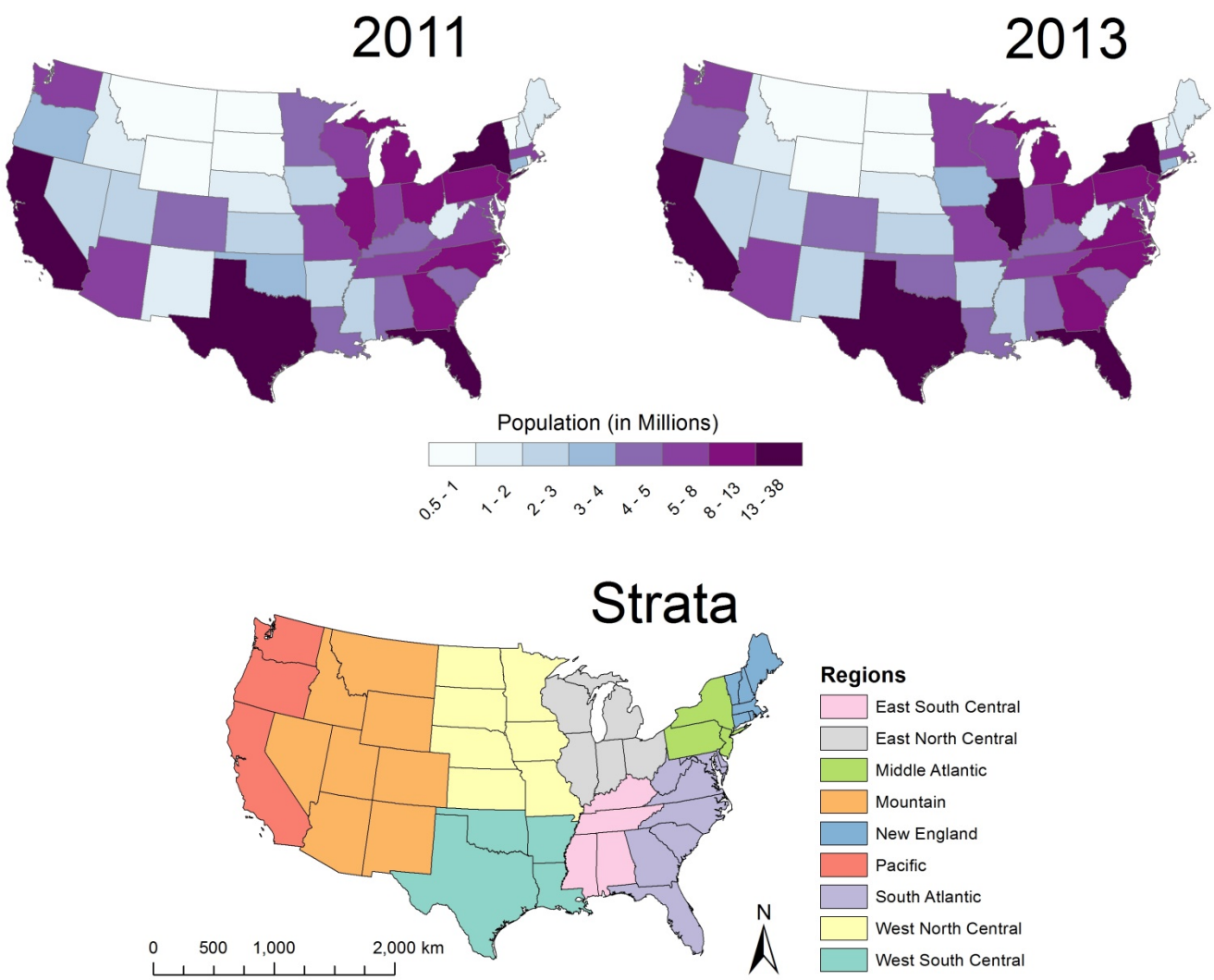

Figure 3. U.S. States shapefile used for testing MAD program. The X field is population in 2010 and the Y field is population in 2013. Each state is an element and each region is a Strata.

Methodology

The script includes three functions which check the projection of the shapefile, project the shapefile if not in equal area and calculates MAD. Before the functions are called, five parameters are initialized. These parameters are user inputs from the GUI (lines 37-42). The next step is where the default datum (WGS_1984) and equal area projection (Sinusoidal (world)) are assigned variables (lines 46-49). A list of the general classes for all equal area projections that ArcGIS supports is under the variable 'prjList' (lines 52-55). This will be used in the first function. A Describe object is created from the input shapefile in order to access the projection information (line 57). A conditional statement is used to determine if the shapefile is unprojected or projected (lines 59-62). This is necessary for the output .csv file.

The first two functions are important to the overall analysis because using MAD for polygons requires weighting the polygons by their area. If the projection of the shapefile is not equal area, the calculation of MAD will be wrong.

The first function 'prjCheck' takes the input shapefile and is used to check if the input shapefile projection class matches any of the equal area projection classes in the list (line 66). If it is an equal area projection, the user receives a message stating "'shapefile name's in an equal area projection. The file will now be used for MAD calculation. Projection is: '+ prjName". The MAD function is then called. If the shapefile is not in an equal area projection, a series of conditional clauses are used to find out if the shapefile can be projected into the default projection system (Sinusoidal). The clauses check the datum to see if it is in the default datum and if it is, the shapefile is projected to the new equal area projection by calling the projection function. If not, the user receives a message stating they must manually project their shapefile into a different datum or equal area projection of their choosing. Another conditional statement gives the message that the shapefile is unprojected if the projection or datum is not assigned to the shapefile. The last conditional statement calls the projection function for shapefiles with the default datum but are unprojected.

The second function 'proj' is called only if the following conditions are met by the shapefile: 1) The shapefile is not in an equal area projection, but has a datum of WGS_1984 2) if the shapefile is unprojected but has a datum of WGS_1984. It is not called if the shapefile is in one of the recognized equal area projection included in prjList.

The third function 'MAD' is called only if the shapefile meets the equal area projection requirements. The first line counts the number of features in the shapefile with is the number of Elements in the shapefile. A new area field is added to the attribute table to make sure the correct areas are used. This is especially important if the shapefile was projected into equal area. The area field is sorted in ascending order and a Search Cursor is applied to find the smallest area polygon in the shapefile. The smallest polygon receives a weight of 1 because it is divided by itself. All other polygon areas are divided by the smallest polygon area and this number is added to another field created called 'weight'. In total, four new fields are added to the shapefile attribute table: 1) area 2) weight 3) weighted deviation 4) absolute weighted deviation. The different values for each of these fields for each polygon in the shapefile are calculated in lines 128-136. Lines 139-157 calculate the mean absolute deviation based on the equation featured in Figure 1.

For the user's convenience, an output .csv file is created with the following information: Input Shapefile, Date Processed, Input Projection, Output Projection, X Field Used, Y Field Used, Stratum Field Used, Total, Quantity, Stratum, and Element. An example output can be seen in Figure 4.

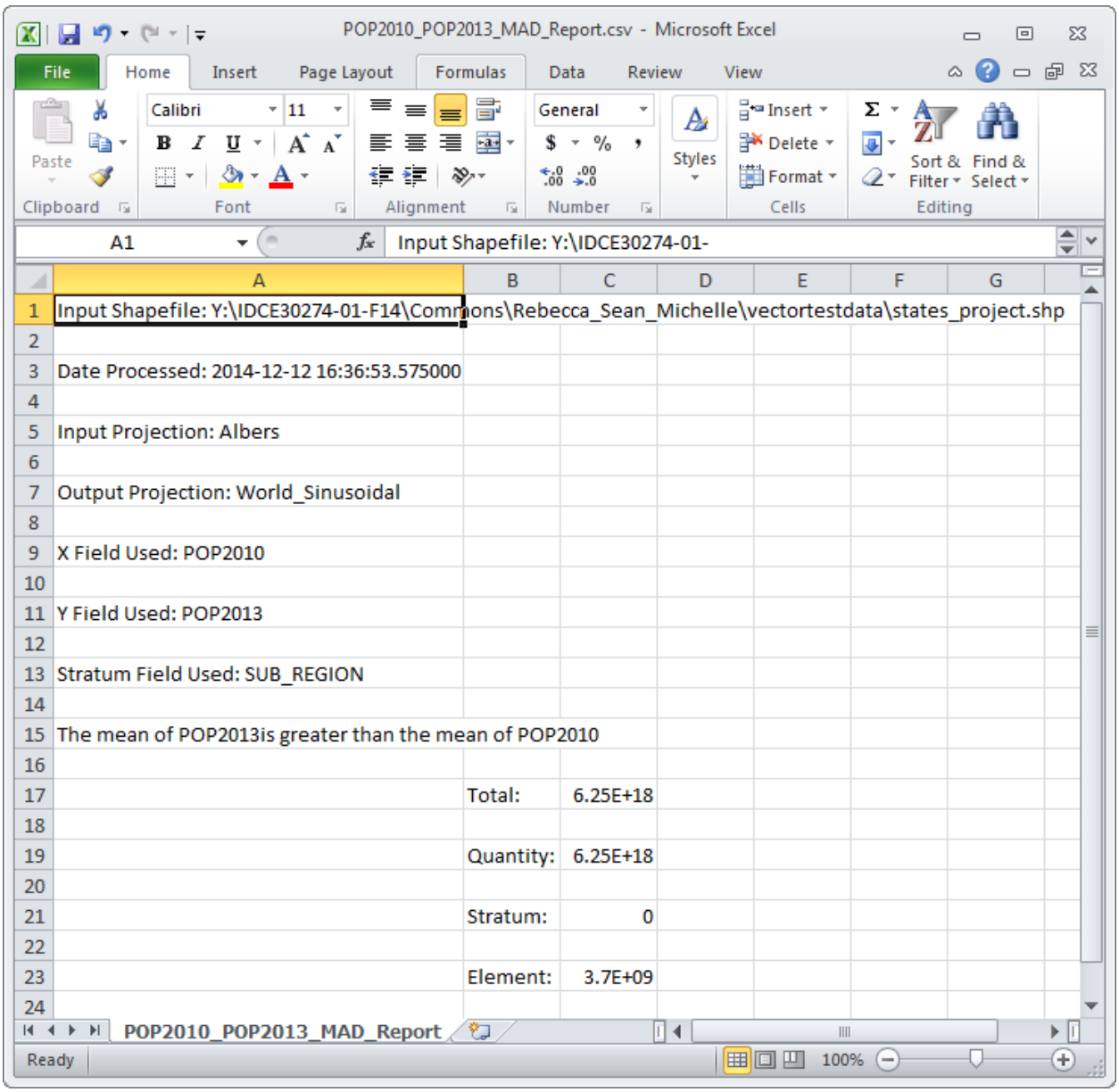

Figure 4. Output comma separated report opened in Microsoft Excel. The report displays the input shapefile, date processed, input projection, output projection, X and Y fields, stratum field, which absolute mean is greater, and the resulting Total, Quantity, Stratum and Element components.

Future modifications to the script could include an optional GUI input where the user decides the equal area projection they would like their areas to be calculated in rather than a default. Another modification would be to include a dictionary of all 5466 projections and their extents to facilitate an automatic datum transformation. Currently the user has to project or transform the datum manually if they get an error message in our script. With this dictionary, the correct datum transformation will be accessed based on the extent, default and current projections of the shapefile.

Discussion

This program permits the broad application of MAD analysis on spatial data. By utilizing the open-source Python programming language, ArcMap and ArcPy toolset this program can be used by a large community of scientist and researchers. Typically in spatial analysis and GIS MAD analysis is conducted using raster datasets; however this use has been modified in order to use vector polygon data. Weighting of data elements is performed using the area of each polygon. The area-weights are correctly calculated by using an equal-area projection. If the input file is not in an equal-area projection the program will automatically re-project it into one.

It is a common misconception in spatial analysis and GIS to use the least square or regression line to compare two datasets of the same variable. Often the coefficient of determination (R^2), which indicates linear covariance between variables two variables, is calculated. However is not appropriate to measure deviation between datasets because R^2 ignores whether the values co-vary. Alternatively, MAD is appropriate because it applies specifically to situations where two variables potentially co-vary. MAD is more relevant than indices that look for agreement, which is difficult to conceptualize and depends on the definition of agreement. MAD's simplicity and focus on deviation makes interpretation easier and contains more information. Applications can include measuring differences between the same variables over time (e.g., population changes), measuring differences between predicted values versus observed values (e.g., model parcel value and real parcel value), and measuring differences attributable to the technique of observation (e.g., different temperature sensors distributed in pairs over space).

Summary

MAD should be computed, reported and interpreted when comparing datasets of the same continuous variable. This program automates MAD analysis within ArcMap GIS between two spatial datasets stored in a single shapefile. A simple to use graphic user interface and automated projection parameters allow for a broad application of this program and MAD analysis. Results of the MAD analysis are output in a comma separated report file that can be opened in Microsoft Excel, facilitating the creation of scatter plots and other descriptive graphics.

Literature

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